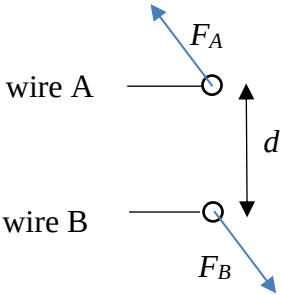


5	
(a)(i)	Out of the page
(a)(ii)1.	$B_E = 35 \mu\text{T}$
(a)(ii)2.	As the d increases (and $1/d$ tends to zero), the magnetic field due to the wire decreases to 0. Hence the y-intercept is only due to the magnetic field due to the Earth.
(a)(ii)3.	From graph, (140, 80) So this implies $B = B_W + B_E$ at this position.
	So $B_W = 80 - 35 = 45 \mu\text{T}$
	Apply $B_w = \frac{\mu_0 I}{2\pi d}$ $I_W = 2\pi\left(\frac{1}{140}\right)(45 \times 10^{-6})(4\pi \times 10^{-7})^{-1} = 1.6 \text{ A}$
(a)(iii)	 Arrows must be labelled in the opposite direction and of the same magnitude. Within 2nd and 4th quadrant.
(b)	Direct current in electromagnet sets up an external magnetic field on the disc
	By Faraday's law, since disc is rotating, there is a rate of change in magnetic flux linkage as the disc enters the region of the external magnetic field.
	This results in an induced emf and a corresponding induced eddy currents in the disc.
	By Lenz's law, these eddy currents will results in a force that oppose the change that is causing it. Hence the velocity of the disc will decrease.

Anglo-Chinese Junior College
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